BY ORDER OF THE COMMANDER
WARNER ROBINS AIR LOGISTICS COMPLEX

WARNER ROBINS AIR LOGISTICS COMPLEX INSTRUCTION 21-111

6 JANUARY 2026
Incorporating Change 1, 5 AUGUST 2026

Maintenance

QUALIFICATION AND CERTIFICATION OF WELDER/WELDER OPERATOR/BRAZER PERSONNEL WELDING AEROSPACE COMPONENTS

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

ACCESSIBILITY: Publication and forms are available on the e-publishing website at www.e-publishing.af.mil for downloading and ordering.

RELEASABILITY: There are no releasability restrictions on this publication.

OPR: 402MXSG

Certified by: 402MXSG/CL (William Broach)

Supersedes: WR-ALCOI21-11, 6 June 2015

Pages: 20

This instruction is the standard written practice for qualification/certification of welder, welder operator, brazer personnel welding aerospace components in Warner Robins Air Logistics Complex (WR-ALC). Technical Order (TO) 00-25-252, *Aeronautical Equipment Welding*, American Welding Society (AWS) D17.1, *Specification for Fusion Welding for Aerospace Applications,* AWS D17.2, *Specification for Welding Aerospace Applications* and AWS B2.2, *Specification for Brazing Procedures and Performance Qualifications* are the controlling documents for this written practice and are made a part of this instruction by reference. This instruction supplements the requirements of the above-mentioned documents to further define procedures as practiced within WR-ALC. In the event of a conflict between the text of this Instruction and TO 00-25-252 and AWS D17.1, the text of above mentioned TOs takes precedence. This instruction applies to welding processes used within WR-ALC. It applies to all personnel welding on aircraft and aircraft support equipment, missile, missile systems, and installed equipment within the 402d Commodities Maintenance Group (402 CMXG), 402d Electronics Maintenance Group (402 EMXG), and 402d Maintenance Support Group (402 MXSG). Refer recommended changes and questions about this publication to the office of primary responsibility (OPR) using Department of the Air Force (DAF) Form 847, *Recommendation for Change of Publication*. This publication may be supplemented at any level, but all direct supplements must

be routed to the OPR of this publication for coordination prior to certification and approval. Requests for waivers must come through the chain of command from the commander or civilian director of the maintenance group or staff office seeking relief from compliance. Waiver requests must be submitted to the OPR; waiver authority has not been delegated. This publication is exempt from tiering pursuant to Department of the Air Force Manual (DAFMAN) 90-161, *Publishing Processes and Procedures.* Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with (IAW) Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and disposed of IAW the Air Force Records Information Management System Records Disposition Schedule located at **https://www.my.af.mil/afrims/afrims/afrims/rds/rds_series.cfm**. See **Attachment 1** for glossary of references and supporting information.

SUMMARY OF CHANGES

This interim change revises WR-ALCI 21-111 by (1) updating visual acuity physical requirements to ensure alignment with higher-level guidance, (2) clarifying the documentation forms for eye examinations, and (3) updating the adopted list of forms. A margin bar (|) indicates newly revised material.

1. GENERAL. 4
2. PROGRAM RESPONSIBILITIES DEFINED. 4
3. TRAINING PROGRAM. 5
4. EXPERIENCE REQUIREMENTS. 5
5. EXAMINATION PRACTICES. 5
6. QUALIFICATION. 5
7. CERTIFICATION. 6
8. FAILED INITIAL CERTIFICATION/RECERTIFICATION ATTEMPT AND DECERTIFICATION. 6
9. RECERTIFICATION. 6
10. RECORDS AND DOCUMENTATION. 6
11. SAFETY. 8

Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION 9

Attachment 2—INSTRUCTIONS FOR COMPLETION OF DD FORM 2757, WELDING EXAMINATION RECORD 13

Attachment 3—INSTRUCTIONS FOR COMPLETION OF WR-ALC FORM 5 WELDER VISUAL ACUITY TEST RECORD (EYE EXAM) 14

Attachment 4—QUALIFICATION OF WELDING PERSONNEL GAS TUNGSTEN ARC WELDING (GTAW) (402 CMXG, 402 EMXG AND 402 MXSG) 15

Attachment 5—QUALIFICATION OF MANUAL BRAZERS (402 CMXG, 402 EMXG, AND 402 MXSG) 17

Attachment 6—QUALIFICATION PROCEDURE FOR RESISTANCE WELDING MACHINE OPERATORS (402 CMXG) 18

Attachment 7—QUALIFICATION OF WELDING PERSONNEL GMAW & SMAW (402 MXSG, 402 CMXG, AND 402 EMXG) 19

1. General.

1.1. **For this document, the term welder will be understood to mean welder/welder operator/brazer unless otherwise stated.**

1.2. **The purpose of welder qualification testing is to determine the ability of the welder to produce a sound weld/brazed joint in the positions that will be encountered in production or support applications.** In addition, the purpose of tests for welder operators is to determine the mechanical ability to operate automated welding equipment.

2. Program Responsibilities Defined.

2.1. **Certifying officials.** The certifying officials and the qualification/certification program management responsibilities, as defined by TO 00-25-252 and AWS D17.1, will reside with the individual welding supervisors within each of the maintenance groups that have certified welders. The certifying officials are designated, in writing, by each group director.

2.1.1. Each group certifying official will be the designated representative (point of contact) for all matters pertaining to welding qualification and certification within that group.

2.1.2. Each group will be responsible for maintaining enough certifying officials to perform these duties.

2.2. **Special Skill Qualification (SSQ) Qualifying Officials.** As required by Air Force Materiel Command Instruction 21-100, Volume 1, *Depot Maintenance Principles*, (AFMCI 21-100V1) SSQ qualifying officials will reside within 402 CMXG and 402 EMXG and be appointed, in writing, by the group commander/director.

2.2.1. SSQ qualifying officials are responsible for performing all applicable functions as specified in AFMCI 21-100V1.

2.2.2. Both 402 CMXG and 402 EMXG will maintain enough SSQ qualifying officials to perform these duties in each organizational group. **Note: 402 MXSG has no SSQ requirements.**

2.3. **Welding supervisors.** Welding supervisors will ensure the candidate completes all prerequisite training and has a current eye exam. Welding supervisors will review all documentation for correctness and discuss the candidate's performance with the SSQ qualifying official or observing official.

2.3.1. Welding supervisors are responsible for maintaining a copy of the completed Department of Defense (DD) Form 2757, *Welding Examination Record* (instructions for completion of DD Form 2757 available at **Attachment 2**), and WR-ALC Form 5, *Welding Visual Acuity Test Record (Eye Exam)* (instructions for completion of WR-ALC Form 5 available at **Attachment 3**), in the welder's training, personnel, or other readily accessible file.

2.4. **Welders.** Welders are responsible for ensuring the proper identification information is correct on the weld/braze specimen materials and ensuring that the welding/brazing procedures are performed IAW with specific welding procedure specifications (WPS) and this instruction.

3. Training Program.

3.1. **SSQ qualifying officials will complete the Air Force Material Command Structured On-the-Job Training (SOJT) trainer course #CTEMAS0000500SU, SSQ officials' course #CHPMAO0000400SU.** The SSQ qualifying official must be qualified and certified in all tasks for which he/she is appointed to qualify welders.

3.2. **Welders must pass the SOJT requirements, if applicable, which pertain to the certification to be obtained.** 402 MXSG welders must complete on-the-job training (OJT) requirements, which pertain to the certification to be obtained.

4. Experience Requirements.

4.1. **Individual skill and talent will determine the time it takes to become a qualified/certified welder.** The minimum number of hours for OJT will not be less than what is required in the appropriate SOJT package.

4.1.1. The SSQ qualifying official (or observing official for 402 MXSG) administering the training will advise the supervisor when he/she considers the candidate ready for qualification/certification.

4.1.2. The welding supervisor will make the final decision on when the candidate is ready to take the qualification/certification test.

4.1.3. The welding supervisor will certify the welder in the Training Scheduling System Production Acceptance Certification/Electronic Training Records (TSS-PAC/ETR) upon successful completion of the proficiency demonstration and the qualification test.

5. Examination Practices.

5.1. **Examinations to verify the physical and technical qualifications of candidates will consist of a physical examination (visual acuity eye examination) and a practical proficiency examination.**

5.1.1. For visual acuity examination physical requirements, refer to T.O. 00-25-252, WP 005 00, **para 5**.

5.1.2. The visual acuity eye exam will be documented on WR-ALC Form 5, in addition to DD Form 2757 and optionally, DD Form 2758, *Welding Certification Card*.

5.1.3. Practical proficiency examination: the practical examination will consist of a demonstration of proficiency by the candidate on tasks that are typical of those to be accomplished in the performance of his/her duties. The unique task requirements are described in Attachments **4 through 7**.

6. Qualification.

6.1. **All personnel assigned to perform welding/brazing on aircraft, missile, engine, accessory, or any other aerospace components will be qualified prior to performing any production welding/brazing.**

6.2. **Qualification procedures, joint requirements, and unique process requirements are described in Attachments 4 through 7, TO 00-25-252 and America Welding Society Standard.**

7. Certification.

7.1. **All personnel assigned to perform welding/brazing on aircraft, missiles, engine, accessory, or any other aerospace components will be certified after the qualification process and prior to performing any production welding/brazing.**

7.1.1. 402 MXSG personnel assigned to weld on aircraft support equipment will be certified after the qualification process and prior to welding.

7.2. **Certification packages are covered in Attachments 4 through 7.**

7.3. **Certification maintenance.**

7.3.1. Welding supervisors are responsible for making sure employees receive the training necessary to enable performance of all tasks within their area of responsibility. Supervisors will review TSS-PAC/ETR automated listings for accuracy and completeness and ensure employees are scheduled for requalification prior to reaching “overdue" status. They also will rotate job assignments to ensure each employee maintains proficiency in each welding/brazing process.

7.3.2. Maintaining proficiency will be documented via AFTO Form 1, *Welding Proficiency Log*, at least every 180 days.

8. Failed Initial Certification/Recertification Attempt And Decertification.

8.1. **Failed initial certification or recertification attempt is covered in Attachments 4 through 7.**

8.2. **Decertification for each welding/brazing process is covered in Attachments 4 through 7.**

9. Recertification. Recertification instructions for each welding/brazing process are described in Attachments **4 through 7**.

10. Records And Documentation.

10.1. **Welder.** The welder is responsible for the completion of the required documentation in Attachments **4 through 7**.

10.2. **SSQ qualifying officials.** SSQ qualifying officials are responsible for the completion of the required documentation in Attachments **4 through 7.**

10.2.1. SSQ qualifying officials will provide the complete package to the welding supervisor for review and brief the welding supervisor on the performance of the welder/brazer and the qualification test results.

10.3. **Welding supervisors.** Welding supervisors will:

10.3.1. Review all documentation required by Attachments **4 through 7**.

10.3.1.1. If the tests are successful, complete the certification package, and execute the certification process described above.

10.3.1.2. If the tests are unsuccessful, refer to Attachments **4 through 7** for further action.

10.3.2. After the group certifying official has signed DD Form 2757, maintain a current copy on file until the next certification cycle.

10.3.3. Maintain a current copy of WR-ALC Form 5 on file until the next visual acuity eye exam has been accomplished. This file must be readily available for audit purposes.

10.4. **Group Certifying Officials.** Group Certifying Officials will:

10.4.1. Review the welder's complete package, fill out block 20, and sign and date block 21 of DD Form 2757.

10.4.1.1. After the completion of **paragraph 10.4.1**, forward the completed SSQ Worksheet to the appropriate 402 CMXG/402 EMXG training/PAC office for input into Section II of the welder's TSS-PAC/ETR record.

10.4.2. Maintain the official certification folder for each welder currently certified to weld/braze within WR-ALC. The official certification folder will be maintained by the certifying official and will contain:

10.4.2.1. WR-ALC Form 5

10.4.2.2. SOJT Worksheet (for each group or process certified if applicable)

10.4.2.3. SSQ Checklist Initial Qualification (for each group or process certified)

10.4.2.4. SSQ Checklist Recertification (for each group or process certified)

10.4.2.5. DD Form 2757 (covering each group or process certified)

10.4.2.6. Specific forms identified in Attachments **4 through 7,** as applicable (see below):

10.4.2.6.1. WR-ALC Form 4, *Gas Tungsten Arc Welding (GTAW) Visual Inspection Report*

10.4.2.6.2. WR-ALC Form 1, *Rate Gyro Case Welder Operator Inspection/Test Record*

10.4.2.6.3. WR-ALC Form 2, *Manual Silver Brazer's Inspection/Test Record*

10.4.2.6.4. WR-ALC Form 3, *Resistance Welder Operator Inspection/Test Record*

10.4.2.6.5. WR-ALC Form 7, *GMAW/SMAW Visual Inspection Report/Test Record*

10.5. **Group training/PAC office.** The group training/PAC office will maintain a copy of the completed SSQ Worksheet for each welder currently certified with the date matching that in Section II of the TSS-PAC/ETR system.

10.5.1. TO 00-25-252 states the certifying facility will maintain records of training and certification/recertification of all qualified welders for the duration of their use (certification cycle). When the certifying official completes the weld package, he/she will forward a copy of the completed SSQ Worksheet to the appropriate group training/PAC manager (see **paragraph 10.4.1.1**.). The supporting documentation will be maintained in the official certification folder.

10.5.2. If requested, a copy of the following optional records may be kept by the welding supervisor to be maintained in the training record. The official date of training completion, TSS-PAC/ETR Section II, must be the date on the SSQ Worksheet, SOJT Worksheet (if applicable), Initial Qualification Worksheet, Recertification Worksheet, and WR-ALC Form 5.

11. Safety.

11.1. **Personal Protective Equipment (PPE).** All welding personnel will be required to wear proper PPE while performing qualification procedures per DAFMAN 91-203, *Air Force Occupational Safety Fire and Health Standards.*

11.2. **Health Risk Assessment.** All welding personnel are required to follow the guidelines set forth in the current health risk assessment (also known as "bio survey") specific to the work area. including the utilization of local exhaust ventilation as required.

DAVID S. MILLER
Brigadier General
Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 33-322, *Records Management and Information Governance Program*, 23 March 2020

AFMCI 21-100V1, *Depot Maintenance Principals,* 22 January 2024

AFSCI 20-114, *Qualification of Nondestructive Inspection Personnel*, 25 January 2024

AWS B2.2, *Specification for Brazing Procedure and Performance Qualification*, 5 August 2016

AWS D17.1, *Specification for Fusion Welding for Aerospace,* 25 March 2024

AWS D17.2/D17.2M:2019, *Specification for Resistance Welding for Aerospace Applications*, 28 February 2018

DAFI 90-160, *Publications and Forms Management*, 14 April 2022

DAFMAN 91-203, *Air Force Occupational Safety*, *Fire, and Health Standards*, 25 March 2022

National Aerospace Standard (NAS) 410, *NAS Certification & Qualification of Nondestructive Test Personnel*, September 2020

TO 00-25-252, *Aeronautical Equipment Welding*, 15 Aug 2022

Prescribed Forms

WR-ALC Form 1, *Rate Gyro Case Welder Operator Inspection/Test Record*

WR-ALC Form 2, *Manual Silver Brazer's Inspection/Test Record*

WR-ALC Form 3, *Resistance Welder Operator Inspection/Test Record*

WR-ALC Form 4, *Gas Tungsten Arc Welding (GTAW) Visual Inspection Report*

WR-ALC Form 5, *Welder Visual Acuity Test Record (Eye Exam)*

WR-ALC Form 7, *GMAW/SMAW Visual Inspection Report/Test Record*

Adopted Form

AF Form 1151, *Training Attendance and Rating*

AFTO Form 1, *Welding Proficiency Log*

DAF Form 847, *Recommendation for Change of Publication*

DD Form 2757, *Welding Examination Record*

DD Form 2758, *Welding Certification Card*

Acronyms

AFI—Air Force Instruction

AFMC—Air Force Materiel Command

AWS—American Welding Society

CMXG—Commodities Maintenance Group

DAFMAN—Department of the Air Force Manual

DD—Department of Defense

EMXG—Electronics Maintenance Group

GMAW—Gas Metal Arc Welding

GTAW—Gas Tungsten Arc Welding

IAW—In Accordance With

MXSG—Maintenance Support Group

OI—Operating Instruction

OJT—On-the-job Training

OPR—Office of Primary Responsibility

PPE—Personal Protective Equipment

RSW—Resistance Spot Welding

SAT—Satisfactory

SMAW—Shielded Metal Arc Welding

SOJT—Structured On-the-Job Training

SSQ—Special Skill Qualification

TO—Technical Order

TSS-PAC/ETR—Training Scheduling System Production Acceptance Certification/Electronic Training Records

UNSAT—Unsatisfactory

WPS—Welding Procedure Specification

WR-ALC—Warner Robins Air Logistics Complex

Terms

Brazer—An individual who performs a manual or semiautomatic brazing operation with a torch.

Brazing—A group of welding processes that produce coalescence of materials by heating them to the brazing temperature in the presence of a filler metal having a liquidus above 840°F (450°C) and below the solidus of the base metal. The filler metal is distributed between the closely fitted faying surfaces of the joint by capillary action.

Certification—The act of determining, verifying, and attesting in writing to the qualification of personnel in conformance to specified requirements.

Certifying Official—According to T.O. 00-25-252, the certifying official will be the examining depot facility, the WR-ALC/CC, or designated representative who certifies the brazers, welders, and welder operators.

Decertification—The act of removing the certification status of an individual, which prevents that individual from performing the task(s) until satisfactory retraining and proficiency demonstration has been accomplished.

Gas Metal Arc Welding (GMAW)—An arc welding process that uses an arc between a continuous filler metal electrode and the weld pool. The process is used with shielding from an externally supplied gas and without the application of pressure.

Gas Tungsten Arc Welding (GTAW)—An arc welding process that produces coalescence of metals by heating them with an arc between a tungsten electrode (non-consumable) and the work pieces. Shielding is obtained from gas. Pressure may or may not be used and filler metal may or may not be used.

Machine Welding—Welding with equipment that performs the welding operation under the constant observation and control of a welder operator. The equipment may or may not load and unload the work pieces.

Nondestructive Inspection—The act of determining the suitability of material or component for its intended purpose using techniques that do not affect its serviceability.

Peel Test or Pull Test—A mechanical test in which the corners (or sides) of seam or spot-welded foil or sheet members are gripped and pulled apart to determine if the joint fails by delamination, by fracture of a cleaved surface, or by tearing of a button (plug) out of the parent material. Delamination, in contrast to fracture, is a sign of no fusion between the two adjacent members. A button or plug, which is pulled out of the removed member, is not necessarily equal in size to the cast nugget underlying it.

Production Acceptance Certification (PAC) Program—Certifies employees to perform and accept completion of assigned work. PAC does this through systematic training, qualification, and certification of individuals.

Qualification—The documented evidence that an individual has completed required training, testing, proficiency demonstration, and possesses the required knowledge and skills to perform a function.

Qualified—Having complied with specific requirements.

Radiography—A permanent visible image on a recording medium produced by penetrating radiation passing through the material being tested.

Recertification—The formal act of requalification and documenting the performance of an individual to the level necessary to satisfy established certification criteria.

Requalification—The act of verifying knowledge and proficiency by testing and demonstrated abilities on a skill or specific group of tasks.

Resistance Welding—A group of welding processes that produces coalescence of the faying surfaces with the heat obtained from resistance of the work to the flow of the welding current in a circuit of which the work is a part and by the application of pressure.

Shielded Metal Arc Welding (SMAW)—Also known as manual metal arc welding, flux shielded arc welding, or informally as stick welding, is a manual arc welding process that uses a consumable electrode coated in flux to lay the weld. An electric current in the form of either alternating or direct current from a welding power supply is used to form an electric arc between the electrode and the metals to be joined. As the weld is laid, the flux coating of the electrode disintegrates, giving off vapors that serve as a shielding gas and providing a layer of slag, both of which protect the weld area from atmospheric contamination.

Special Skill Qualification (SSQ)—Skills so specialized they require extensive technical knowledge and proficiency. SSQ is a prerequisite to PAC task certification.

SSQ Disqualification—The act of removing the SSQ qualification status of an individual when requalification requirements are not maintained. PAC task decertification follows SSQ disqualification.

SSQ Proficiency Checklist—Identifies the standards and step-by-step documentation of SSQ qualification/requalification requirements.

SSQ Qualifying Official—An individual authorized in writing by the group commander/director to witness and evaluate skill qualifications of welding candidates using appropriate SSQ guides/checklists.

SSQ Requalification—The act of verifying SSQ knowledge and proficiency by testing and demonstrating abilities on a skill or specific group of tasks.

Weld—A localized coalescence of metals or nonmetals produced either by heating the materials to the welding temperature, with or without the application of pressure, or by the application of pressure alone, with or without the use of filler metal.

Welder—An individual who performs a manual or semiautomatic welding operation, when used in this document, refers to welder, welder operator and brazer.

Welder Operator—and **Welding Operator -** One who operates adaptive control, automatic, mechanized, or robotic welding equipment. This includes spot welding machines.

Attachment 2

INSTRUCTIONS FOR COMPLETION OF DD FORM 2757, WELDING EXAMINATION RECORD

Figure A2.1. Instructions For Completion of DD Form 2757, Welding Examination Record.

(INSTRUCTIONS ARE SPECIFIC FOR WR-ALC)

SECTION I

Block 1 APPLICANT NAME: Name of candidate being certified

Block 2 RANK/GRADE: Welder's/welder operator's/brazer's pay grade (i.e., WG-7/8/9/10/11)

Block 3 WELDER IDENTIFICATION: Welder's identification number

Block 4 ORGANIZATION/UNIT ADDRESS: Candidate's unit and unit address

Block 5 WELDING SCHOOL/SOURCE OF TRAINING: 402 CMXG, 402 EMXG, or 402 MXSG

Block 6 DATE OF LAST EYE EXAM: Date of last eye exam

Block 7 QUALIFICATION GROUP: Metal groups submitted for certification

Block 8 WELDING PROCESS: Whatever process certifying (i.e., GTAW or Torch Brazing or Resistance Welding

Block 9 WELDING PROCEDURE USED: WPS Number from TO 00-25-252 (Note: Resistance Spot Welding enter "RSW")

Block 10 SIGNATURE OF APPLICANT: Self-explanatory

Block 11 SIGNATURE OF OBSERVING OFFICIAL: Signature of the SSQ official who observed the certification weld

SECTION II

Block 12 WPS #: Enter WPS number from block 9.

Block 13/14 TEST/RETEST SPECIMEN: Will be marked either satisfactory (SAT) or unsatisfactory (UNSAT), IAW the results of the examination, by the SSQ official who observed the certification weld. (The SSQ official will write the test type in block 15, then sign adjacent to it).

Block 15 REMARKS: In addition to the signatures, the SSQ official will record the cause of any unsatisfactory conditions.

Block 18 STATUS OF APPLICANT: select Qualified or Not Qualified.

Block 19 SIGNATURE OF TEST OFFICIAL: Apply signature.

Block 20 ORGANIZATION AND ADDRESS: Organization (i.e., 402 CMXG, 402 EMXG, or 402 MXSG).

Block 21 SIGNATURE OF CERTIFYING OFFICIAL: Group certifying official who certifies the welder/ welder operator/brazer will enter signature and record the effective date of certification.

Attachment 3

INSTRUCTIONS FOR COMPLETION OF WR-ALC FORM 5 WELDER VISUAL ACUITY TEST RECORD (EYE EXAM)

Figure A3.1. Instructions For Completion of WR-ALC Form 5 Welder Visual Acuity Test Record (Eye Exam).

The welder/brazer will hand carry the form to the Occupational Medical Clinic in building 207 and present the form to medical personnel administrating the eye exam. The medical personnel will complete the vision examination portion of the form using the guidance given in the upper portion of this form. The welder/brazer will return the form to his/her supervisor. If the results are acceptable, the supervisor will keep a copy for the welder's/brazer's shop file and forward a copy to the group certifying official for the official certification folder. If the results do not meet the requirements of AWS D17.1, AWS D17.2, AWS B2.2, and T.O. WP 005 02. The welder/brazer will not be allowed to perform welding/brazing until vision has been corrected by visual aids and the welder/brazer has passed the eye exam. Once the welder/brazer has passed the eye exam, the welder/brazer's supervisor will maintain a copy of the acceptable form and forward a copy to the group certifying official (if different from supervisor) for inclusion into the official certification folder.

Attachment 4

QUALIFICATION OF WELDING PERSONNEL GAS TUNGSTEN ARC WELDING (GTAW) (402 CMXG, 402 EMXG AND 402 MXSG)

A4.1. PURPOSE.

A4.1.1. A directive is required to specify the technical details of qualification testing of welding personnel for certification/recertification required IAW TO 00-25-252.

A4.2. SCOPE. See TO 00-25-252 for requirements on GTAW certification.

A4.2.1. CMXG and EMXG welders will be required to have Level II certification in GTAW using the 6G and 3F coupon for each metal group required.

A4.2.2. MXSG welders will be required to have Level I certification in GTAW using 2G, 3G, and 3F coupons.

A4.2.3. The welding supervisor will ensure the candidate has completed all prerequisite training and has a current eye exam. The welding supervisor will review all documentation for correctness and discuss the candidate's performance with the SSQ qualifying official.

A4.2.4. SSQ qualifying official training requirements. (No required SSQ's for 402 MXSG)

A4.2.5. Must complete both the AFMC SOJT trainer course #CTEMAS0000500SU and SSQ officials' course #CHPMAO0000400SU.

A4.2.6. Must pass the appropriate SOJT course(s) applicable to the metal groups the candidate is seeking qualification.

A4.2.7. Must be appointed in writing.

A4.2.8. Welding candidate training requirements. Welding candidates seeking certification must pass the appropriate SOJT training (if applicable) prior to being tested for qualification if applicable.

Figure A4.1. Documentation.

Documentation

Required weld package for initial qualification:

SOJT, Training Work Sheet (if applicable)

SSQ Qualification Checklist (for each group) No required SSQ's for 402 MXSG

WR-ALC Form 5 (current)

WR-ALC Form 4 (for each group)

DD Form 2757 (covering all groups)

Weld specimen

Required associated documentation for requalification:
SSQ Qualification Checklist (for each group) No required SSQ's for 402 MXSG

WR-ALC Form 5 (current)

WR-ALC Form 4 (for each group)

DD Form 2757 (covering all groups)

Weld specimen

Required associated documentation listed below for requalification after failed attempt to requalify:

SOJT, Training Work Sheet (if applicable)

SSQ Qualification Checklist (for each group) No required SSQ's for 402 MXSG

WR-ALC Form 5 (current)

WR-ALC Form 4 (for each group)

DD Form 2757 (covering all groups)

Weld specimen

Attachment 5

QUALIFICATION OF MANUAL BRAZERS (402 CMXG, 402 EMXG, AND 402 MXSG)

A5.1. PURPOSE.

A5.1.1. A directive is required to specify the technical details of qualification testing of manual brazers for certification/recertification required IAW TO 00-25-252.

A5.2. SCOPE.

A5.2.1. See TO 00-25-252 WP005 for required certification.

A5.2.2. No required SSQ's for 402 MXSG

Figure A5.1. Documentation.

Documentation

Required brazer package for initial qualification:

SOJT, Training Work Sheet (if applicable)

SSQ Qualification Checklist (No required SSQ's for 402 MXSG)

WR-ALC Form 5 (current)

WR-ALC Form 2

DD Form 2757

Silver braze specimens

Required associated documentation for requalification:

SSQ Qualification Checklist (No required SSQs for 402 MXSG)

WR-ALC Form 5 (current)

WR-ALC Form 2

DD Form 2757

Silver braze specimens

Required associated documentation for requalification after failed attempt to requalify:

SOJT Training Work Sheet (if applicable)

SSQ Qualification Checklist (No required SSQ's for 402 MXSG)

WR-ALC Form 5 (current)

WR-ALC Form 2

DD Form 2757

Silver braze specimens

Attachment 6

QUALIFICATION PROCEDURE FOR RESISTANCE WELDING MACHINE OPERATORS (402 CMXG)

A6.1. PURPOSE.

A6.1.1. A directive is required to specify the technical details of qualification testing of resistance welder operator for certification/recertification required IAW TO 00-25-252.

A6.2. SCOPE.

A6.2.1. See TO 00-25-252, section 32, for the required certifications process. The qualification period for this process is 1 year.

Figure A6.1. Documentation.

Documentation

Required weld package for initial qualification:

SOJT, Training Work Sheet (if applicable)

SSQ Qualification Checklist

WR-ALC Form 5 (current)

WR-ALC Form 3

DD Form 2757

For a satisfactory RSW performance test, produce 10 welds: Five (5) each of 0.063 inch and Five (5) each of 0.032 inch.

Required associated documentation listed below for requalification:

SSQ Qualification Checklist

WR-ALC Form 5 (current)

WR-ALC Form 3

DD Form 2757

For a satisfactory RSW performance test, produce 10 welds: Five (5) each of 0.063 inch and Five (5) each of 0.032 inch.

Required associated documentation listed below for requalification after failed attempt to requalify:

SOJT, Training Work Sheet (if applicable)

SSQ Qualification Checklist

WR-ALC Form 5 (current)

WR-ALC Form 3

DD Form 2757

For a satisfactory RSW performance test, produce 10 welds: Five (5) each of 0.063 inch and Five (5) each of 0.032 inch.

Attachment 7

QUALIFICATION OF WELDING PERSONNEL GMAW & SMAW (402 MXSG, 402 CMXG, AND 402 EMXG)

A7.1. PURPOSE.

A7.1.1. A directive is required to specify the technical details of qualification testing of welding personnel for certification/recertification required IAW TO 00-25-252.

A7.1.2. No SSQ will be required for 402 MXSG.

A7.2. QUALIFICATION PROCEDURES. See TO 00-25-252 for GMAW and SMAW certification requirements.

A7.2.1. CMXG and EMXG welders will be required to have certification in GMAW & SMAW using the 1G and 2F coupon for each metal group required.

A7.2.2. **MXSG welders will be required to have certification in SMAW using the 6G, with 4" pipe coupon for each metal group required.**

A7.2.3. **MXSG welders will be required to have certification in GMAW using the 2G, 3G, and 3F sheet coupons.**

A7.2.4. Welding candidate training requirements. Welding candidates seeking certification must pass the appropriate SOJT training (if applicable) prior to being tested for qualification.

Figure A7.1. Documentation.

Documentation

Required weld package for initial qualification:

WR-ALC Form 5 (current)

WR-ALC Form 7

DD Form 2757 (covering all groups)

Weld specimen

Required associated documentation for requalification:

WR-ALC Form 5 (current)

WR-ALC Form 7

DD Form 2757 (covering all groups)

Weld specimen

Required associated documentation listed below for requalification after failed attempt to requalify:

WR-ALC Form 5 (current)

WR-ALC Form 7

DD Form 2757 (covering all groups)

Weld specimen